Compiling GNU Software for Windows

Let's find out how exactly a cross-platform software is compiled for Windows, and then let's do it ourselves!

Now there's this great software called The GIMP, which is more than a match for Photoshop, and it's smaller, and it's free! The GIMP, although made primarily for GNU systems tacked on top of a UNIX kernel (such as Linux), is also available for Windows. Ditto for the Apache Web server. Ever wondered how they write the source code once, and then build it for any operating system they choose? Ever wondered how *you* could do it for yourself?

Welcome to GCC

First of all, of course, you must write code that is platform independent, i.e., that uses instructions that can be reproduced in any operating system. As long as you write software using functions defined in the standard C library, minus POSIX functions (such as *fork()*), you are fine. It'll build. You can go to a GNU machine and pass your file through GCC to get a Linux binary, then go to a Windows box and pass it through VC++ to get a Windows binary.

Now, as you move to large programs (such as The GIMP), you will need multiple source code files to produce a single binary. This will result in problems, because you cannot just build the files at one go. You will need to compile your C sources into the assembler, then assemble them into object files, put all the object files together in a single object archive and then link that archive to the system libraries. The answer to this problem is *Makefiles*, which can do all that automatically according to a set of rules applied to a set of files. But even *Makefiles* have their limitations, and *Makefiles* written for GNU Make are not compatible with Microsoft NMAKE and a *Makefile* written to use GCC cannot use VC++, and vice versa.

The answer to this problem is to use something called a ***retargetable compiler***, or a compiler that can produce binaries for different operating systems. Fortunately for us, GCC is retargetable. And unfortunately for us, GCC must be re-built before it can produce Win32 or Win64 code.